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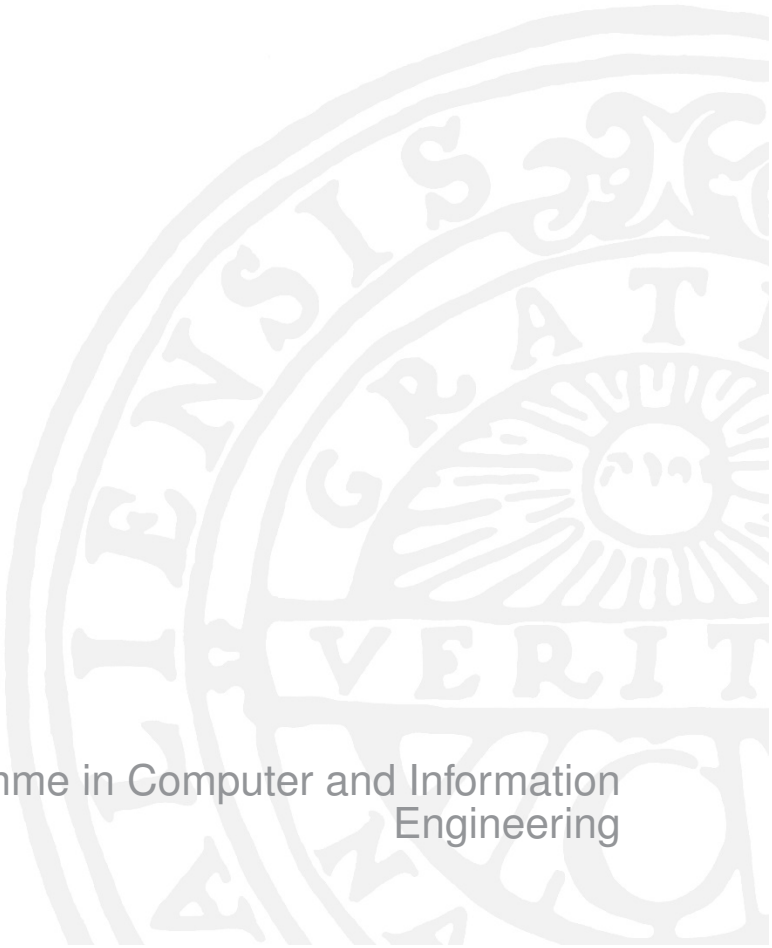
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Designing for Informal Social Interaction in Hybrid Work

A Study of a 2D Virtual Workspace

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Master's Programme in Computer and Information Engineering



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Abstract

The shift toward hybrid and remote work has increased reliance on digital tools for collaboration. However, most platforms support structured, task-oriented communication while providing limited opportunities for spontaneous and informal interaction. This thesis explores how a 2D virtual workspace can be designed to support informal social interaction in hybrid work settings. A user-centered design approach was applied in collaboration with employees at Knowit Connectivity. The study combined interviews, thematic analysis, iterative prototype development, and think-aloud evaluations. The interviews revealed five recurring themes. These highlighted the importance of informal interaction for workplace relationships, the increased threshold for interaction in hybrid work, the role of visibility and social presence, the value of shared spaces and activities, and the importance of usability for adoption. Based on these themes, a virtual lounge prototype was designed and implemented. The prototype included presence visibility, availability status indicators, shared voice chat, and mini-games. Evaluation of the high-fidelity prototype showed that participants experienced the environment as more socially natural and less transactional than conventional tools such as Microsoft Teams. The study demonstrates that supporting informal interaction in hybrid work is not only a matter of providing communication tools. It also requires creating social approachability, where colleagues feel present, reachable, and open to interaction with minimal effort. The findings further show that a user-centered design approach can help translate these needs into meaningful design features. Overall, the results suggest that supporting informal interaction digitally requires a shift in focus from task efficiency toward social atmosphere and low-threshold engagement. The findings also suggest that further development and research are needed to understand how platforms of this type can be integrated into existing work practices and provide long-term value for organizations.

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Sammanfattning

Övergången till hybrid- och distansarbete har lett till en ökad användning av digitala samarbetsverktyg, men de flesta plattformar är främst utformade för strukturerad och uppgiftsorienterad kommunikation och erbjuder begränsade möjligheter till spontan och informell interaktion. Detta examensarbete undersöker hur en 2D-virtuell arbetsyta kan utformas för att stödja informell social interaktion i hybridarbete genom en användarcentrerad designmetod, i samarbete med anställda på Knowit Connectivity. Intervjuer genomfördes för att identifiera sociala och interaktiva behov, följt av iterativ prototyputveckling och utvärdering genom användbarhetstestning med tänka högt-metoden. Analysen av intervjuerna resulterade i fem centrala teman: informell interaktions betydelse för arbetsrelationer, svårigheten att initiera spontan kontakt i hybridarbete, vikten av synlighet och social närvaro, värdet av delade miljöer och aktiviteter samt användbarhetens betydelse för plattformsanvändning. Utifrån dessa teman utformades och implementerades en virtuell loungeprototyp med funktioner för närvarosynlighet, tillgänglighetsstatus, delad röstchatt och gemensamma minispel. Utvärderingen visade att deltagarna upplevde miljön som socialt mer naturlig och mindre transaktionell än konventionella verktyg som Microsoft Teams. Studien bidrar med insikten att informell interaktion i hybridarbete inte enbart handlar om tillgång till kommunikationsverktyg, utan även om social tillgänglighet. Det behöver vara enkelt att se vilka kollegor som är närvarande, tillgängliga och öppna för spontana samtal. Resultaten visar också att användarcentrerad design kan bidra till att omvandla identifierade behov till meningsfulla designfunktioner som stödjer social interaktion i hybridarbete. Att stödja informell interaktion digitalt kräver därför ett annat förhållningssätt än traditionella samarbetsverktyg. Istället för att prioritera effektivitet och strukturerade arbetsflöden behöver utformningen främja social atmosfär och en låg tröskel för interaktion. Resultaten visar samtidigt att fortsatt utveckling och forskning behövs för att förstå hur plattformar av denna typ kan integreras i befintliga arbetssätt och skapa långsiktigt värde i organisationer.

Acknowledgement

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1 Introduction

The way people work and collaborate has changed significantly in recent years. Following the COVID-19 pandemic, many organizations transitioned to hybrid and remote work models, increasing their reliance on digital technologies for communication and collaboration [11, 19].

While collaborative digital tools effectively support task execution, coordination, and structured communication, they do not fully support the kinds of social interaction that take place in physical work settings. Research shows that these tools provide limited support for spontaneous and informal social interaction [38]. This type of interaction occurs naturally in physical workplaces and plays an important role in building relationships, strengthening team unity, and supporting organizational culture [21, 27].

The limitations of current digital communication tools become particularly evident when opportunities for informal interaction are reduced. In physical offices, casual conversations and unplanned encounters occur naturally and facilitate communication and coordination among coworkers [27]. In remote work settings, the frequency of scheduled meetings and other real-time interactions decreases, while informal and spontaneous exchanges decline, resulting in more fragmented and less connected collaboration networks [39]. To address these challenges, recent research has explored spatial platforms such as Gather.town and their ability to facilitate informal interaction through proximity-based communication and visual presence [1]. However, limited attention has been devoted to the design and evaluation of digital environments within the context of established hybrid organizations. For organizations seeking to maintain social connections in digitally mediated work, this constitutes a significant research gap.

1.1 Purpose

Against this background, there is a need to better understand how digital environments can be designed to support informal social interaction in hybrid work settings. The purpose of this thesis is therefore to explore how a digital environment can be designed to support informal social interaction in hybrid work. Specifically, the study examines the design and evaluation of a 2D virtual workspace. The workspace is intended to function as a dedicated social space that supports informal interaction among coworkers. Rather than serving as another productivity tool, the environment aims to provide a place where employees can engage in spontaneous conversations and social interaction alongside their regular work tasks. To address this purpose, the following research questions guide the study:

RQ: How can a 2D virtual workspace be designed to support informal social interaction in hybrid work settings?

The study is guided by the following sub-questions:

- What social and interaction-related needs do employees experience in hybrid and remote work?
- How do employees experience the developed prototype in relation to identified social and collaborative challenges?

2 Background

This chapter presents the theoretical background of the study. It begins by examining the role of informal interaction in work and the psychological need for social connection, drawing on Self-Determination Theory. It then outlines the shift toward hybrid work and the challenges this creates for spontaneous and informal communication. The chapter further explores the concepts of social presence and awareness in digital work, before reviewing existing digital environments and their capacity to support informal interaction. Finally, perspectives on user experience and work engagement in digital environments are discussed. Together, these sections provide the foundation for understanding the design challenge addressed in this thesis and motivate the study's research questions.

2.1 The Role of Informal Interaction in Work

Informal interaction in a work setting refers to spontaneous and unstructured exchanges that occur outside formally defined tasks or meetings [6]. Examples include casual conversations, brief check-ins, and other unplanned exchanges between colleagues. Informal interactions matter because they offer opportunities for connection beyond what structured meetings and/or task-based communication typically provide. When people work together in an office setting, informal interactions occur frequently due to the proximity of coworkers and the unplanned interactions that result from sharing a common physical space [27]. This could help with relationship building and support coordination among team members [27].

Research on virtual organizations shows that informal socialization remains important even in distributed teams. For example, Cimperman [6] has demonstrated that despite

geographical separation, informal socialization continues to remain a key element of maintaining connection and reducing feelings of isolation among remote workers. Similarly, Koch *et al.* [21] demonstrate that informal communication is positively related to employees' sense of being informed, affective commitment, job satisfaction, and productivity. Therefore, informal interaction supports both social connection and organizational outcomes [6, 21]. Limiting opportunities for such interactions in hybrid and remote work settings may therefore have consequences that extend beyond social discomfort. Taken together, these studies suggest that limiting opportunities for informal interaction in hybrid and remote work settings risks weakening both employees' sense of organizational belonging and their overall engagement at work.

Given the importance of social interaction for employees' experiences at work, it is relevant to consider underlying psychological mechanisms. One prominent theory addressing this is the Self-Determination Theory (SDT), which provides a framework for understanding human motivation and well-being [28]. Within this theory, Deci and Ryan [28] propose that individuals have three basic psychological needs: autonomy, competence, and relatedness. For this study, it is mainly relatedness that is of relevance and the concept refers to the desire to feel connected to others, to experience a sense of belonging, and to maintain meaningful social relationships. Research shows that supportive relationships and meaningful interaction promote autonomous motivation and positive work outcomes [9, 28], while environments that limit opportunities for connection may reduce feelings of belonging and undermine motivation [9].

In remote and hybrid work settings, the conditions for fulfilling the need for relatedness are altered. As interaction becomes more dependent on digital tools, opportunities for spontaneous social contact decrease, making it more difficult to maintain social connections. When the need for relatedness is not met, employees may experience decreased motivation, lowered engagement, and a weaker feeling of belonging [9]. This highlights that when designing a digital environment, it should not only focus on providing an environment that supports collaboration on task-related objectives, but also enabling social interaction and connection among colleagues.

2.2 The Shift Toward Hybrid Work

Remote and technology-based collaborative environments have long been studied within the field of Computer-Supported Cooperative Work (CSCW). Olson and Olson [27] demonstrate that while technology enables coordination of activities over distance, distributed work has its own social and organizational hurdles that are less visible when working together in the same space. Specifically, problems with shared context and mutual understanding have become major concerns in distributed collaboration [7]. Cram-

ton [7] illustrates that because geographically dispersed teams are unable to share or interpret contextual information, it can lead to misunderstandings, create incorrect assumptions regarding their co-workers' behavior, and may reduce the overall unity of the team.

Although these challenges had been discussed for many years, remote work remained small in scale until the COVID-19 pandemic. The crisis led to a rapid, large-scale shift from office-based to home-based work, accelerating existing trends toward digitally enabled collaboration across organizations [16, 19]. This was later followed by more stable hybrid work arrangements [16]. With work becoming fully mediated by information and communication technologies (ICTs), everyday interaction increasingly relies on digital tools rather than face-to-face communication [38]. Digital tools enable teams to continue collaborating at a distance, however, they create additional obstacles to communication quality and coordination [38]. Maintaining effective teamwork often requires more structured processes and conscious effort than in traditional work settings [38].

Previous research on distributed collaboration [7, 27] revealed that there are many long-standing challenges associated with remote and distributed work. As hybrid work has become increasingly common across many organizations [16, 19], these challenges are no longer temporary or situational, they represent an ongoing condition in many work settings. The natural, informal interactions and socialization that occur in physical offices are now taking place in digital environments. Often, digital environments do not facilitate the informal social elements of work, creating a barrier to employees developing relationships with colleagues and feeling part of a community, all important factors contributing to employee motivation and well-being.

2.3 Challenges to Informal Interaction in Hybrid Work

The shift toward remote and hybrid work changes the social conditions of collaboration. Research shows that co-located teams benefit from informal "hallway conversations" and spontaneous exchanges that arise through shared physical presence [27]. These interactions are often considered important for supporting collaboration, social relationships, and shared understanding in the workplace [21, 27]. However, in distributed settings, communication technologies can fall short in supporting the richness and spontaneity of face-to-face interaction, making informal exchanges less frequent and harder to initiate [27].

Remote work also affects how awareness and transparency are experienced within teams, where transparency refers to keeping track of colleagues' activities, progress, and re-

sponsibilities. While digital tools can support visibility through status indicators and shared documents, transparency is shaped not only by technology but also by team norms and organizational structures [4]. Increased visibility in digital tools may support coordination, but it can also create feelings of exposure or surveillance [36]. In remote settings, awareness therefore becomes more structured and tool-based, rather than naturally emerging through physical presence at the workplace.

In addition, remote work has been linked to more fragmented communication networks and fewer spontaneous cross-team interactions [39]. Remote work often relies on asynchronous communication channels, such as email, as well as formal scheduled meetings [39]. Although these communication methods support coordination, they can slow down communication and limit opportunities for spontaneous conversation, thereby creating barriers to informal exchange [39]. These findings indicate that both hybrid and remote work reduce the probability of employees' spontaneous interactions with each other, and therefore their common knowledge. Communication becomes much more planned and focused on tasks, and less time is available for non-task-related socializing. Therefore, there are now many lost opportunities for employees to build relationships, coordinate actions, and experience an "in-common" sense of community.

2.4 Social Presence and Awareness in Digital Work

One useful concept for understanding why informal interaction becomes less frequent and harder to initiate at a distance is social presence, defined as the feeling of being "with" another real person in a mediated environment [26]. When social presence is strong, colleagues feel socially available to each other despite physical distance [26], making interaction feel more natural. When presence signals are limited, informal exchanges become harder to initiate, meaning that the spontaneous interaction that occurs naturally in co-located offices is more difficult to recreate digitally. The link between social presence and informal interaction becomes particularly visible in spatial digital environments. Aoki *et al.* [1] found that users of Gather.town experienced a stronger sense of co-presence compared to traditional videoconferencing tools. The proximity-based interaction of the platform supported more spontaneous exchanges, similar to those occurring in co-located offices. At the same time, the study identified usability challenges and organizational factors that affected continued use. These findings suggest that spatial and visual cues can support informal interaction in hybrid work, but that careful design is needed to balance social features with usability and everyday work practices. This suggests that if such a balance is not achieved, the platform may risk being abandoned or used primarily for formal tasks, rather than supporting the informal interaction it was intended to enable.

2.5 Virtual Workspaces and Digital Platforms

Digital collaboration in hybrid work is commonly supported by platforms such as Microsoft Teams, Zoom, and Slack [23, 32, 40]. These tools organize communication through scheduled meetings, messaging channels, and shared documents. Project management systems such as Jira and Trello further support task coordination [17, 34]. Although digital tools for collaboration provide an important function when working from home or remote locations, most of the time, they are used primarily for structured interactions related to specific work projects. In addition to the structured use of digital tools, communication in distributed work environments often requires more deliberate and effortful interaction processes than face-to-face communication [7]. This can hinder informal communication and make it more difficult to develop shared understanding among team members.

As hybrid work has become more common, new types of digital environments have emerged to address challenges such as reduced spontaneous interaction and social isolation [1]. Spatial platforms such as Gather.Town and Kumospace provide shared digital environments where users are represented by avatars and can move freely in a virtual space. Communication is based on proximity, which means that audio or video interaction is activated when users approach each other. This allows conversations to happen more spontaneously compared to traditional meeting-based platforms. These environments aim to recreate certain aspects of informal interaction found in physical offices [1].

Research has examined how such spatial environments affect collaboration, and Sánchez *et al.* [30] found that desktop-based spatial environments supported collaboration at levels similar to in-person and videoconferencing settings. However, immersive virtual reality increased cognitive load. This suggests that higher levels of immersion do not automatically improve collaboration and that usability remains important. Similarly, Aoki *et al.* [1] showed that Gather.town increased users' sense of presence by enabling proximity-based "walk-up" conversations. However, the study also identified several practical challenges related to usability and long-term adoption. Their study focused on fully remote software teams and highlights the need to explore how such environments function in different organizational contexts. By examining a hybrid work setting, this study extends this line of research to contexts where employees alternate between remote and co-located work.

Overall, these studies suggest that virtual spatial environments can support informal interaction in hybrid work. However, their success depends not only on creating a sense of presence but also on maintaining usability and fitting into existing work practices.

2.6 UX and Design Considerations

As work becomes increasingly mediated through digital tools, understanding how these environments are experienced by users becomes important. This perspective is commonly captured through the concept of user experience (UX). According to ISO 9241-210, UX refers to a person's perceptions and responses that result from using a system, product, or service [15]. UX is influenced both by system design and by the context in which the system is used. In workplace settings, this means that UX concerns how digital tools shape employees' daily work, collaboration, and overall experience, not only how efficiently tasks are completed. Simsek Caglar *et al.* [31] emphasize that research on UX in work contexts remains limited and show that many studies focus mainly on usability and efficiency, treating UX as equal to functionality. Broader experiential aspects, such as enjoyment, motivation, and emotional engagement, are less frequently discussed. This narrow focus makes it difficult to understand how digital systems affect employees' overall work experience. Accordingly, there is a need to examine not only whether the systems function properly but also how they are experienced by users.

Another way to understand the experiential aspects of digital work environments is through the concept of work engagement. Bakker *et al.* [2] describe work engagement as a positive state characterized by energy, dedication, and involvement. It is influenced not only by task demands but also by job resources such as autonomy, social support, and opportunities for interaction. From a design perspective, this highlights the importance of creating digital environments that support social interaction, as these factors contribute to employees' overall experience and engagement.

Recent research shows how digital collaboration tools can influence both UX and work engagement. Simsek Caglar *et al.* [5] found that real-time collaboration in an online whiteboard can create a sense of a shared "digital office." The participants' ability to see each other's interactions, along with receiving immediate feedback from others, was associated with a stronger sense of social presence among participants. The ease and playfulness of the features provided for users to participate was directly related to participants' perceptions of autonomy, competence, and relatedness. The work by Aoki *et al.* [1] further shows that UX extends beyond functionality. Several factors were found to affect continued use of the digital platform Gather.town. These included sensory overload, physical discomfort caused by prolonged headphone use, and concerns that its playful aesthetic was inappropriate for a professional setting. These studies suggest that digital environments not only support tasks but also shape employees' sense of connection and involvement. They also show that both usability and broader experiential qualities matter for whether a platform is actually used.

2.7 Theoretical Synthesis

In recent years, the nature of the workplace has changed dramatically. With technological advancements enabling many workers to perform their jobs remotely or from hybrid workplaces, collaboration at a distance using technology has become much more common. While digital tools enable collaboration across distance, they also change the conditions for social interaction. Many workplace tools primarily support structured and task-oriented communication, which can make spontaneous and informal interaction more difficult to initiate. This is important since informal interaction has been shown to support relationship-building, coordination, and social connection among coworkers [21, 27]. Remote workers will have to put extra effort into initiating unplanned conversations compared to those who are physically present in the workplace. The Social Presence theory provides insight into why unplanned conversation may seem less natural in mediated environments. Previous research also links informal interaction and relatedness to job satisfaction, motivation, engagement, and workplace belonging [6, 9, 21, 28].

From a design perspective, this suggests that digital work environments should provide support for both both task-oriented collaboration and informal social interaction. Research on spatial digital environments has found that platform designs that include features such as proximity-based interaction and visible presence provide greater opportunities for spontaneous communication [1, 30]. However, less is known about how such environments can be designed and evaluated within established hybrid organizations. Existing studies, such as Aoki *et al.* [1] focus on fully remote teams and do not address the specific challenges of hybrid work settings, where employees alternate between working remotely and in the office. This suggests a gap in the literature, and one that this thesis directly addresses by designing and evaluating a 2D virtual workspace intended to support informal social interaction in a hybrid work setting. Accordingly, this study contributes with new knowledge on both the design of such environments and how employees experience them in relation to their social and collaborative needs.

3 Method

The overall aim of this study is to examine how a 2D virtual workspace can support informal social interaction in hybrid work settings. To address this, the study was divided into a three-phase design process, where each phase addressed different aspects of the research questions. Phase 1 explored employees' social and interaction-related needs in hybrid and remote work, Phase 2 focused on the development and usability evaluation of a prototype, and Phase 3 examined how employees experience the developed

prototype in relation to the identified social and collaborative challenges. A qualitative approach, grounded in User-Centered Design, was considered appropriate for the study.

3.1 Case Description

This study was conducted as a case study in collaboration with Knowit Connectivity, one of the business areas within the IT consultancy firm Knowit. Knowit is a knowledge-intensive organization with approximately 3900 employees, operating across several domains related to digital transformation, including software development, cloud services, and embedded systems [20]. Given the organization's strong reliance on digital collaboration and established hybrid work practices, it provides a relevant context for studying informal social interaction in distributed work environments. As a consultancy organization, employees at Knowit Connectivity primarily work in project-based teams that are often distributed across locations and client assignments. As a result, daily work is highly dependent on digital tools for communication, coordination, and knowledge sharing. Commonly used platforms at Knowit Connectivity include Microsoft Teams and Jira, which effectively support structured and task-oriented collaboration. However, these tools provide limited support for informal and spontaneous social interactions, such as casual conversations or unplanned encounters, which are otherwise common in co-located office environments.

In this project, employees at Knowit Connectivity are the user group. The participants included roles such as software developers, scrum masters, project managers, and consultant managers, with varied experience of hybrid work. These users were involved throughout the research process in accordance with User-Centered Design principles, contributing to both the identification of user needs and the evaluation of the product. Conducting this study within Knowit provided access to a real-world context, creating a deeper understanding of employees' everyday work practices and the challenges associated with informal interaction in a hybrid work setting.

3.2 User-Centered Design

User-Centered Design (UCD) is an approach to system design that aims to improve usability by focusing on users, their needs, and their interaction with a system [15]. It emphasizes a clear understanding of users, tasks, and the context of use. It also promotes continuous user involvement throughout an iterative process of design and evaluation [15]. UCD is particularly relevant for this study, as it focuses on understanding and supporting informal social interaction in hybrid work environments, which is

closely related to the experiences and perceptions of users. Norman [25] highlights that effective design must be grounded in user needs, behaviors, and mental models, and that usability problems often arise when systems do not align with user expectations. Therefore, involving users throughout the design and evaluation process is essential for developing meaningful and usable solutions.

In addition to these general principles, the ISO 9241-210 standard describes UCD as an iterative process consisting of four key activities: understanding the context of use, specifying user requirements, producing design solutions, and evaluating the designs against requirements [14]. These activities are revisited and repeated throughout the design process as new insights appear. In this study, this iterative process was applied, and the following sections describe how each of these activities was carried out. See Figure 1 for an overview of the process.

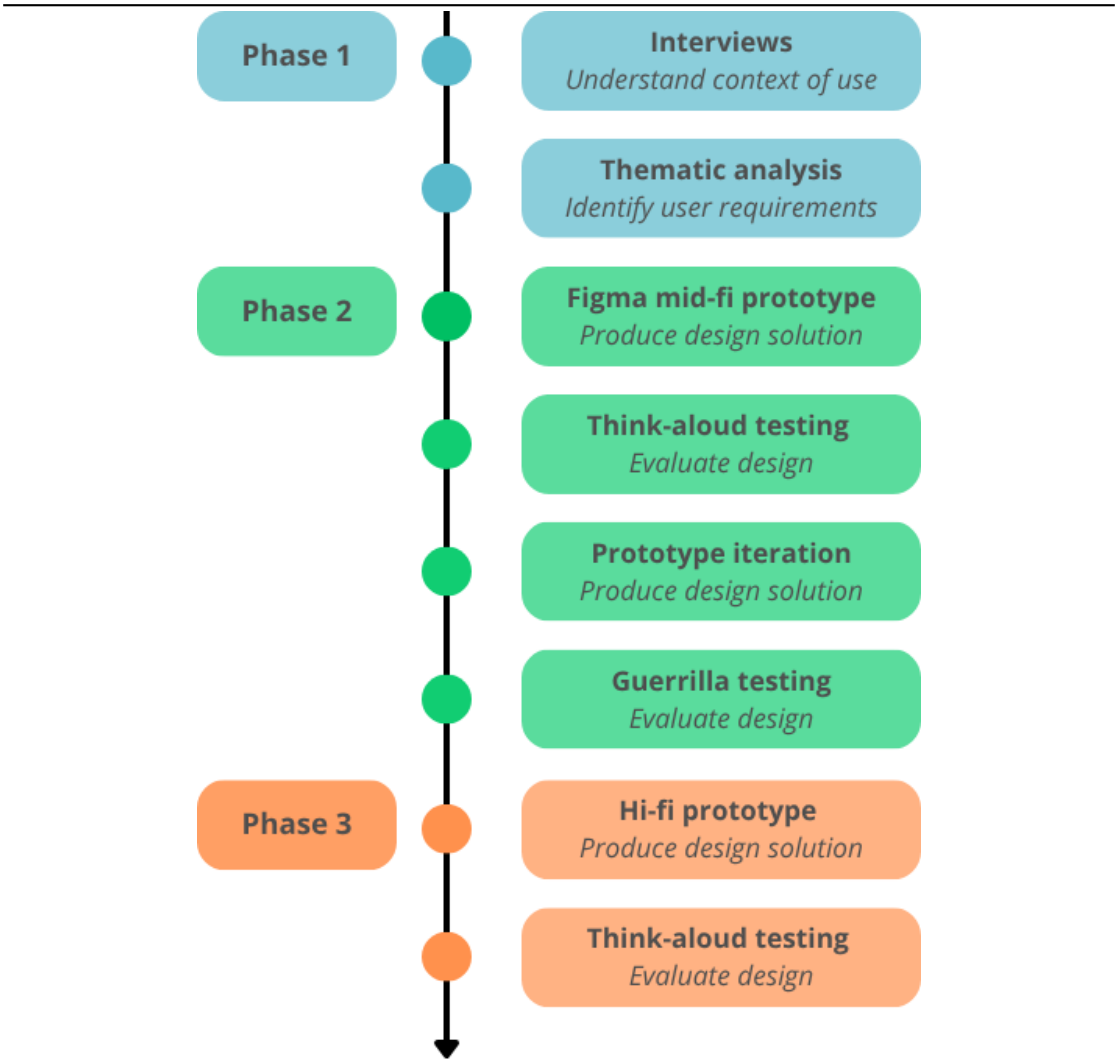


Figure 1 Overview of the study process, structured as an iterative design workflow inspired by user-centered design principles and methods [14].

3.2.1 Phase 1: Understanding Context of Use & Identifying User Requirements

Phase 1 aimed to explore employees' social and collegial needs in hybrid and remote work. Understanding how informal interaction functions in physical office environments and how it changes in digital settings was considered essential for identifying what a digital environment needs to support.

A qualitative research approach was chosen because the study aims to understand the

experiences, perceptions and social interactions of employees in hybrid work. This approach was appropriate for exploring complex social experiences in hybrid work settings, where a more detailed understanding was needed. According to Creswell *et al.* [8], qualitative research is appropriate when a problem needs to be thoroughly explored and when numerical data alone cannot fully explain it. Although surveys could have provided measurable data on interaction frequency or satisfaction, they would not have offered the same depth of understanding. Therefore, a qualitative approach was considered more suitable for exploring informal social interaction in hybrid work.

3.2.1.1 Interview Protocol and Procedure

Semi-structured interviews were used to collect data in this phase of the project. This method allows the researcher to guide the conversation while giving participants the freedom to describe their experiences in their own words [29]. The interview guide was developed according to the research objectives and was based on the principles outlined in Saldaña's *Fundamentals of Qualitative Research* [29]. The book informed both the structure of the guide and the conduct of the interviews, including the use of open-ended questions, active listening, and follow-up prompts to deepen the responses of the participants.

The guide ensured that the interviews covered relevant themes, including informal interaction, social presence, perceived availability, and challenges in digital communication. The guide provided structure while allowing participants to introduce additional relevant experiences. Examples of questions in the interview guide include: "What role does informal interaction play in your work life?", "When you think about working at the office versus working remotely, what feels different socially?", and "What do you think such a platform should include to feel meaningful and worth using for you?" See Appendix A.1 for the full interview guide.

A pilot interview was conducted before the main data collection to test the clarity and order of the questions, as well as the overall duration of the interview, following recommendations to test the protocol before the main study [29]. The interviews were conducted in accordance with established research ethics principles, including the requirements for information and informed consent [37]. Before each interview, participants were informed about the purpose of the study, how the data would be handled, and that participation was voluntary, and their consent was obtained before proceeding.

In total, seven semi-structured interviews were conducted, with a duration of around 40 minutes each. This number was sufficient to identify recurring themes related to the social and collegial needs of employees in hybrid work environments. The interviews were conducted in Swedish or English, depending on the language preferred by the par-

ticipant. The participants were employees at Knowit Connectivity, primarily working as software developers, with a smaller number of consultant managers. This composition provided perspectives from both technical and managerial roles within the organization.

3.2.1.2 Data Analysis

Braun and Clarke's [3] six-step guide was used to analyze the interview data through a thematic analysis. This method was chosen because it allows the identification of meaningful patterns in a dataset in relation to specific research questions [3]. Step 1 involved familiarization with the data. All interviews were audio-recorded and transcribed using AI-assisted transcription software [35]. The transcripts were then carefully reviewed and manually corrected while listening to the original recordings to ensure accuracy. The interviews were conducted in Swedish or English, and the transcripts were kept in their original language during the analysis to preserve meaning. As recommended by Braun and Clarke [3], the transcripts were re-read and the recordings revisited to increase familiarity with the data. This process helped initial ideas and observations to emerge.

In Step 2, the initial codes of the transcripts were generated. The transcripts were systematically reviewed to identify segments of text relevant to the research questions. These included descriptions of informal interaction in the office, differences between office and remote work, barriers to informal interaction, and suggestions for how a digital platform could support such interaction. Relevant quotes were collected and labeled with short descriptive codes. These coded extracts and their associated codes were then organized into a code-book to support the following steps of the analysis.

Step 3 focused on finding potential themes. The codes from the previous step were examined and compared in order to identify patterns across the dataset. A thematic map was created to explore relationships between the codes. Codes that described similar ideas were grouped together into clusters, which were then used to develop candidate themes representing broader patterns in the data [3]. The next step focused on reviewing and refining the themes. In Step 4, the candidate themes were examined in relation to the coded extracts to determine whether they formed clear patterns. Some themes were merged when they described similar ideas. For example, themes relating to the positive role of informal interaction and spontaneous interaction in the office were combined, as both described how informal interaction occurs naturally in shared spaces. Similarly, themes related to missing interaction in hybrid work and barriers to digital communication were combined into a broader theme that describes challenges to informal interaction in hybrid work settings. The themes were then checked against the entire dataset to ensure that they reflected the patterns found across the interviews [3].

Step 5 focused on defining and naming the themes more clearly [3]. During this stage, the meaning and scope of each theme were clarified by reviewing the coded extracts. Each theme was then given a clear and descriptive name. Finally, Step 6 consisted of writing up the analysis. At this stage, the themes were organized and presented in relation to the research questions. Relevant extracts from the interviews were selected to illustrate each theme and translated into English.

3.2.2 Phase 2: Prototype Design and Iterative Evaluation

Based on the themes identified during the thematic analysis, functional requirements for the digital platform were specified. These requirements defined what the system needed to support in order to address the identified user needs. They also guided the development of a prototype that explored how informal social interaction could be supported in a digital environment. A mid-fidelity interactive web prototype was created in Figma. Mid-fidelity prototypes represent a level of detail between low- and high-fidelity, offering more detail than paper sketches while still allowing for rapid iteration [12]. Choosing a mid-fidelity prototype was a practical decision, as the design concept was already reasonably defined and a slightly more detailed representation made it easier to evaluate the social and spatial aspects of the environment. The prototype included basic interactive features while still containing some placeholder elements. Visual assets, such as furniture, avatars, and tileable floor patterns, were generated using an AI-based image generation tool (ChatGPT/DALL·E) and exported as PNG files, which were then integrated into the prototype.

3.2.2.1 Evaluating the Design Against Requirements

To evaluate the prototype's usability, a think-aloud study was conducted with four participants. This method was chosen because it allows observation of how users naturally interact with the interface and helps identify usability issues as they occur. By verbalizing their thoughts, users reveal how they interpret the interface and where misconceptions arise, providing direct insight into which parts of the design cause problems [24]. According to Nielsen [24], thinking-aloud testing is particularly suited for iterative design and formative evaluation, and three to five users are sufficient to identify the most significant usability issues.

The participants were Knowit Connectivity employees, representing the same user group as in the interviews, including consultant managers and software developers. This ensured that the feedback reflected the perspectives of the intended users of the platform. Each session consisted of five tasks described in Table 1, designed to evaluate how in-

tuitively participants could navigate the core features of the interface.

Table 1: Task scenarios used during the evaluation of the mid-fidelity prototype in Phase 2.

Task	Description
1	Create a profile and enter the lounge
2	Find and start a game with a colleague
3	Find a different game to play
4	Change your status from *Available* to *Do Not Disturb*
5	Log out of the lounge

Participants were asked to think aloud throughout each session, verbalizing their thoughts, reactions, and reasoning as they worked through the tasks. Following the sessions, a short debrief was conducted where participants were asked open-ended questions about their overall experience with the prototype, such as what they would change or improve and whether anything felt confusing or unexpected. In line with Nielsen's recommendation [22], the researcher refrained from intervening during the sessions in order to preserve natural user behavior and allow usability issues to emerge organically.

After the evaluation, the feedback and observations were analyzed, and improvements were made to the Figma prototype based on the findings. The revised prototype was then briefly evaluated by one user through guerrilla usability testing. Guerrilla usability testing is an informal and lightweight method in which users quickly interact with a design and provide immediate feedback [13]. Even a small number of participants can reveal valuable usability issues, and early testing is often more beneficial than more extensive evaluation conducted later in the process [22]. This evaluation was used to verify that the implemented changes improved the design before proceeding to implementation. This phase primarily addressed the usability and design aspects of the prototype, forming the basis of the subsequent evaluation of the user experience in Phase 3.

3.2.3 Phase 3: System Implementation and Evaluation

This phase directly addressed the second sub-question by examining how users experience the prototype in relation to the identified social and collaborative challenges. The high-fidelity platform was implemented as a web-based system based on the visual design established in Phase 2 and accessed through a web browser during testing. The frontend was developed in React, while the interactive lounge environment was implemented using Phaser 3. The backend was built with Node.js, where Express handled

HTTP endpoints and Colyseus enabled real-time communication between users. This allowed user movements, status updates, and interactions to be synchronized across clients. Voice communication was implemented using LiveKit, and user authentication and persistent data storage were handled through Supabase and PostgreSQL. This implementation enabled a realistic evaluation of how the platform supports informal social interaction in practice.

3.2.3.1 Evaluating User Experience of the Prototype

The high-fidelity prototype evaluation was conducted with four participants, following the same think-aloud methodology as during Phase 2 [24]. Sessions were recorded with permission from the participants. The evaluation combined task-based interaction with a semi-structured post-test interview. This approach was used to explore how users experienced the prototype in relation to the social and collaborative challenges identified in the study. Specifically, it addressed research sub-question 2 (”How do employees experience the developed prototype in relation to identified social and collaborative challenges?”). Each session began with a brief introduction to the study, providing participants with the contextual background needed to reflect meaningfully on their experience. The participants then completed the tasks shown in Table 2.

Table 2: Tasks used during the evaluation of the high-fidelity prototype in Phase 3.

Task	Description
1	Create an account and enter the lounge
2	Explore the lounge and describe what you think you can do there
3	Find a game in the lounge and try to start playing it
4	Change your display name or availability status

To simulate a more realistic social environment, example users were present in the lounge during testing. After completing the structured tasks, participants were encouraged to explore the platform freely. As voice communication could not be tested within the session, it was described to participants so that it could be considered in their responses. The session concluded with a set of post-test questions (see Table 3), each linked to the themes identified in the initial interviews.

Table 3: Post-test interview questions used in the evaluation of the high-fidelity prototype and their relation to the identified interview themes.

ID	Question	Theme(s)
Q1	Does the environment feel like a place where you could naturally "drop in" spontaneously?	1, 2
Q2	Does it feel easy or difficult to approach other users in the environment?	2, 3
Q3	Does initiating contact here feel easier than via tools like Teams?	2
Q4	Do the activities feel like natural social entry points?	4
Q5	Does the visibility of others feel helpful or intrusive?	3
Q6	What would it take for you to actually want to use a platform like this with colleagues?	5

Participants were also asked an open-ended closing question about potential improvements or additional features.

4 Results

This section presents the findings from the three phases of the study. It begins by reporting the themes identified through the thematic analysis of the interviews, describing the social and interaction-related needs employees experience in hybrid and remote work. These themes are then used to derive a set of functional requirements that guided the design of the prototype. The section then presents the prototype across its iterations, from the initial Figma design through usability testing to the final high-fidelity implementation, and concludes with the findings from the high-fidelity prototype evaluation.

4.1 Phase 1: User Needs in Hybrid Work

This section presents the themes identified through the thematic analysis, addressing the research sub-question: "What social and interaction-related needs do employees experience in hybrid and remote work?" The analysis resulted in five themes that describe key aspects of informal interaction in hybrid work environments and how digital environments could support such interaction. These themes are presented in the following sections.

4.1.1 Informal interaction strengthens workplace relationships (Theme 1)

Participants frequently described informal interaction as an important part of everyday work life. Conversations during breaks, lunches, or spontaneous discussions were described as helping colleagues get to know each other and making the workplace more enjoyable. Several participants described how informal conversations contribute to a positive workplace atmosphere. For example, a participant explained:

> "It creates a feeling of... like a feeling of home when you go to the office and meet your colleagues."

Another participant described how informal interaction leads to new social connections with colleagues:

> "You meet new friends and talk about other things during lunch and coffee breaks, not just about work."

Participants also described how these conversations support collaboration and problem solving. Supporting such interaction in digital environments may, therefore, be important, as it can make it easier to ask quick questions or discuss ideas without scheduling meetings. As one participant noted:

> "I see it as highly beneficial for development, and many important things often emerge through informal conversations."

Together, these findings suggest that informal interaction plays both social and practical roles in the workplace. Supporting informal interaction in digital environments may therefore be important for fostering both collaboration and a sense of social connection and belonging. This aligns with the concept of relatedness in Self-Determination Theory [28], as informal interaction enables employees to build and maintain social connections, which are important for supporting motivation and engagement in the workplace.

4.1.2 Hybrid work raises the threshold for informal interaction (Theme 2)

Participants described clear differences between interaction in the office and interaction when working remotely. In the office, informal interaction often occurs naturally when colleagues are physically close to each other whereas such opportunities were described as less frequent in remote settings. The participants explained that communication when working remotely is often limited to work-related issues. As one participant noted:

> "Those random talks almost never happen because if you call someone it's usually something work related."

Several participants also described remote work as more isolating due to the reduced interaction with colleagues:

> "It almost feels like working in isolation at home."

Another challenge mentioned was hesitation when contacting colleagues digitally. Without visual cues indicating whether someone is available, participants often felt uncertain about interrupting others. As one participant explained:

> "The threshold is much higher when you work remotely because you have to send a message and you don't know if the person is busy."

The findings indicate that hybrid work environments increase the effort required to initiate informal conversations. This can be understood in terms of reduced social presence, where the lack of visible cues makes colleagues feel less accessible. As a result, informal communication becomes more effortful and less spontaneous. This points to an important design challenge for a 2D virtual workspace: how to lower the threshold for informal interaction in a hybrid work setting.

4.1.3 Colleague visibility enables spontaneous communication (Theme 3)

Participants described how visibility of colleagues in a digital environment could support spontaneous interaction. In physical workplaces, visual awareness of colleagues often triggers informal conversations. Participants suggested that similar visibility in digital spaces could recreate this effect. Visual representations such as avatars were described as a way of making colleagues feel present and approachable in a digital environment. Participants highlighted the value of being able to see who is in the space and what they are currently doing, since it could lower the threshold for initiating informal interaction. As one participant noted:

> "I would like it [a potential platform] to be a bit more... so that you can see who is there."

Another participant described how visibility of colleagues' availability in a potential digital lounge could encourage informal interaction:

> "If there is a lounge area where people are just sitting... I know that these people are quite free. I can just go there and have a regular conversation."

These findings indicate that visibility plays an important role in enabling informal interaction in digital environments. This can be understood through the concept of social presence, where visible cues of others' availability make colleagues feel more approachable and easier to engage with. From a design perspective, features supporting visibility and awareness therefore appear important for enabling spontaneous interaction in digital work environments.

4.1.4 Digital spaces and activities could support informal interaction (Theme 4)

Participants suggested that digital environments could support informal interaction by providing dedicated spaces and shared activities. Digital lounges and rooms were described as places where colleagues could gather without scheduling meetings, similar to break areas in physical offices. As one participant expressed:

> "Rooms you can jump into where people are discussing different things."

In addition, shared activities such as mini-games were described as a way to create casual moments of interaction in digital environments. These activities were seen as a way of encouraging colleagues to spend time together informally, without the need to schedule or plan the interaction. As another participant noted:

> "The most fun would be if there was some kind of mini-game."

These findings suggest that both informal spaces and lightweight shared activities can support spontaneous interaction in digital work environments. This can be understood as the creation of a shared context, which reduces the effort required to initiate interaction. This interpretation aligns with Olson and Olson's [27] discussion of the importance of shared context in distributed collaboration. From this perspective, digital environments may partially replicate the conditions that enable informal conversations in co-located workplaces. From a design perspective, this indicates that environments should not only make colleagues visible, but also provide shared contexts that naturally invite interaction.

4.1.5 Usability and participation influence platform adoption (Theme 5)

Participants also emphasized that practical factors strongly influence whether a digital platform designed for informal interaction would actually be used. Ease of use and technical performance were mentioned as particularly important. One participant stated:

> "If it is too complicated or not user friendly I would just give up on it."

Similarly, technical reliability was described as a key factor for adoption:

> "If the performance is poor or it lags, people will not use it."

Participants also pointed out that a platform would only be meaningful if colleagues actively participated:

> "If no one is there, it becomes completely pointless to use it"

These findings suggest that even if a digital platform is designed to support informal interaction, it may not be adopted if it is difficult to use or if colleagues do not actively participate. This indicates that usability and active participation are important conditions for a platform to fulfill its intended purpose in an organizational context. From a design perspective, this highlights the importance of creating systems that are both easy to use and capable of attracting and sustaining user engagement within the organization. This aligns with the findings of Aoki *et al.* [1], who identified usability challenges and organizational factors as key barriers to the continued use of a virtual 2D workspace.

4.1.6 Functional Requirements

Based on the themes related to social and interaction-related needs experienced by employees in hybrid and remote work, a set of Functional Requirements (FRs) was specified to guide the design of the prototype. Table 4 presents these requirements and shows how each is grounded in one or more of the identified themes.

The requirements were derived directly from the thematic analysis. FR1 draws on Theme 1, where informal interaction was described as important for relationship-building, and Theme 4, where participants suggested that shared spaces and activities could recreate these interactions in digital environments. FR2 is also rooted in Theme 4, where

participants described how shared activities such as games can provide a natural entry point for informal interaction. Shared activities were perceived as reducing the effort required to initiate contact and making social engagement feel more spontaneous. FR3 draws on Themes 2 and 3. Participants described how working remotely made them hesitant to contact colleagues because they could not tell if someone was busy, while Theme 3 highlighted how awareness of presence and availability could reduce that hesitation. FR4 is based on Themes 1 and 2, capturing the contrast between spontaneous interaction in physical offices and its absence in remote work. The requirement reflects the need to support low-effort, unplanned communication. Finally, FR5 is derived from Theme 5, where participants were clear that a platform should be easy to use and technically reliable. Participants emphasized that systems perceived as complex or unreliable would not be used.

Table 4: Functional requirements derived from the thematic analysis and their relation to identified themes.

ID	Functional Requirement	Theme(s)
FR1	The system shall support informal, non-task-oriented interaction by enabling users to engage in lightweight social activities within a shared environment.	T1, T4
FR2	The system shall support informal interaction by enabling shared activities that give users a natural reason to engage with others without a work-related agenda.	T4
FR3	The system shall make it easy for users to perceive who is present and potentially available for interaction.	T2, T3
FR4	The system shall enable users to initiate interaction without prior scheduling or formal setup.	T1, T2
FR5	The system shall be simple and technically reliable ensuring that usability and performance do not become barriers to participation.	T5

4.2 Phase 2: Prototype Design and Iteration

Based on the functional requirements identified in Phase 1, an initial prototype was developed to explore how informal social interaction could be supported within a shared digital environment. This phase presents the initial design concept together with the usability findings that informed later design refinements.

4.2.1 Initial Mid-Fidelity Prototype

The prototype is a 2D top-down lounge with a 16-bit retro aesthetic, designed to resemble a classic video game rather than a conventional work tool. Visually appealing design has been shown to improve user experience and perceived usability [33], and game design elements can make interactions more enjoyable and less task-focused [10]. The lounge includes two game tables for ping pong and chess, placeholder areas for potential themed rooms, and a sidebar displaying which colleagues are currently present, along with their availability status. Each user is represented by a pixel art avatar visible within the space. Figure 2 shows an overview of the prototype.

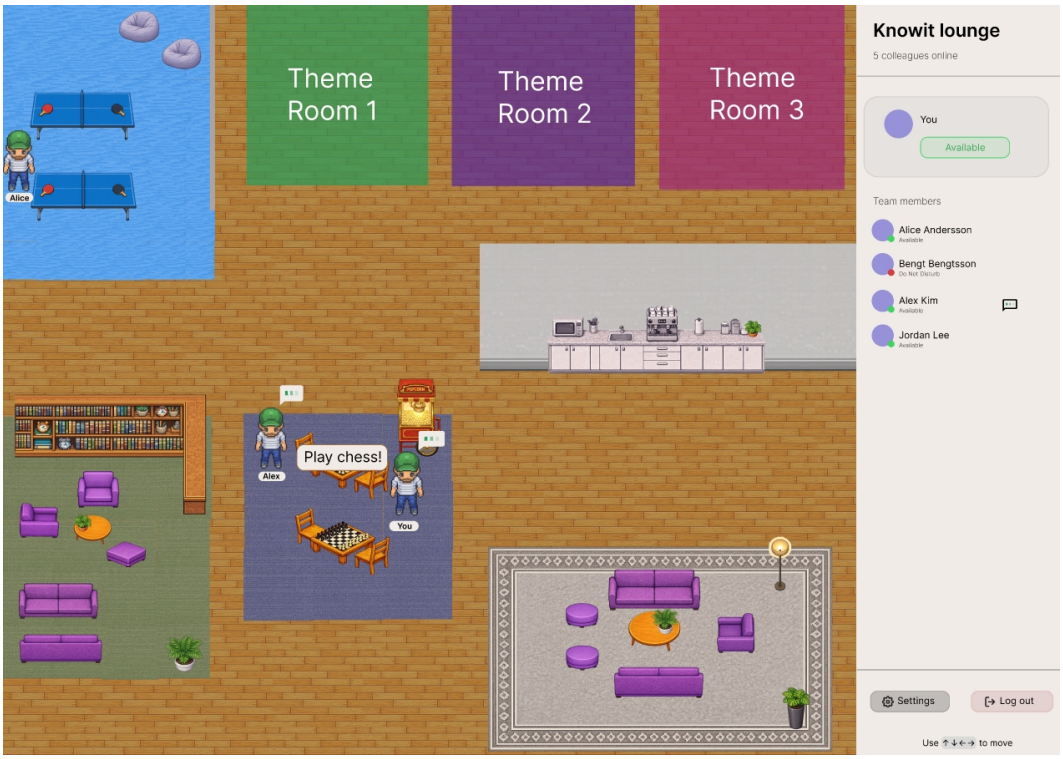


Figure 2 Overview of the first prototype iteration.

The design of the prototype reflects the functional requirements identified in Section 4.1.6. The informal and playful atmosphere of the lounge directly supports FR1, by encouraging non-task-oriented interaction within a shared environment. The game tables address FR2 by giving users a natural reason to spend time in the space, allowing informal communication to happen without requiring explicit initiation. Visibility and awareness are supported through avatars and the sidebar, addressing FR3 by making it easy to see who is present and whether they are available, lowering the threshold for

initiating informal interaction. FR4 is represented by the ability to simply approach another avatar, shown in the Figma version as a speech bubble with three dots above two avatars positioned close to each other. The interface was kept simple and easy to navigate throughout, reflecting FR5 and ensuring that the design itself does not become a barrier to participation.

4.2.2 Findings from the Think-Aloud Evaluation

The think-aloud evaluation provided insights into how participants interacted with the prototype and revealed several usability-related issues and areas for improvement. All four participants in the user evaluation successfully completed every task. Tasks 1, 4, and 5 (see Table 1) were completed quickly and without hesitation, indicating that creating a profile, changing status, and logging out were intuitive interactions. The most consistent difficulties occurred in Tasks 2 and 3, which involved finding and starting games. Once the correct interaction was discovered, the participants continued without further difficulty.

Several usability issues were observed across sessions. The icon used to represent active voice was frequently misinterpreted as a text or chat indicator, with participants generally suggesting that a microphone icon would be more intuitive. In addition, a recurring need for clearer onboarding and in-lounge guidance was identified. While only one participant explicitly suggested adding contextual tips or hints, similar moments of uncertainty were observed across all sessions. This suggests that clearer guidance would benefit the overall user experience. Overall, the sessions indicated that the core flows were discoverable and functional, but that onboarding and in-lounge clarity could be improved.

4.2.3 Refined Mid-Fidelity Prototype

In the second iteration of the prototype (see Figure 3), several usability improvements were implemented based on the findings from the think-aloud evaluation. A tip box was added to improve onboarding, and the speech bubble was updated with a microphone icon to clarify voice interaction. Other smaller adjustments were also made to improve overall usability, such as adding microphone and sound buttons so users can quickly change their audio settings. These changes aimed to improve clarity and reduce interaction uncertainty identified in the initial evaluation.

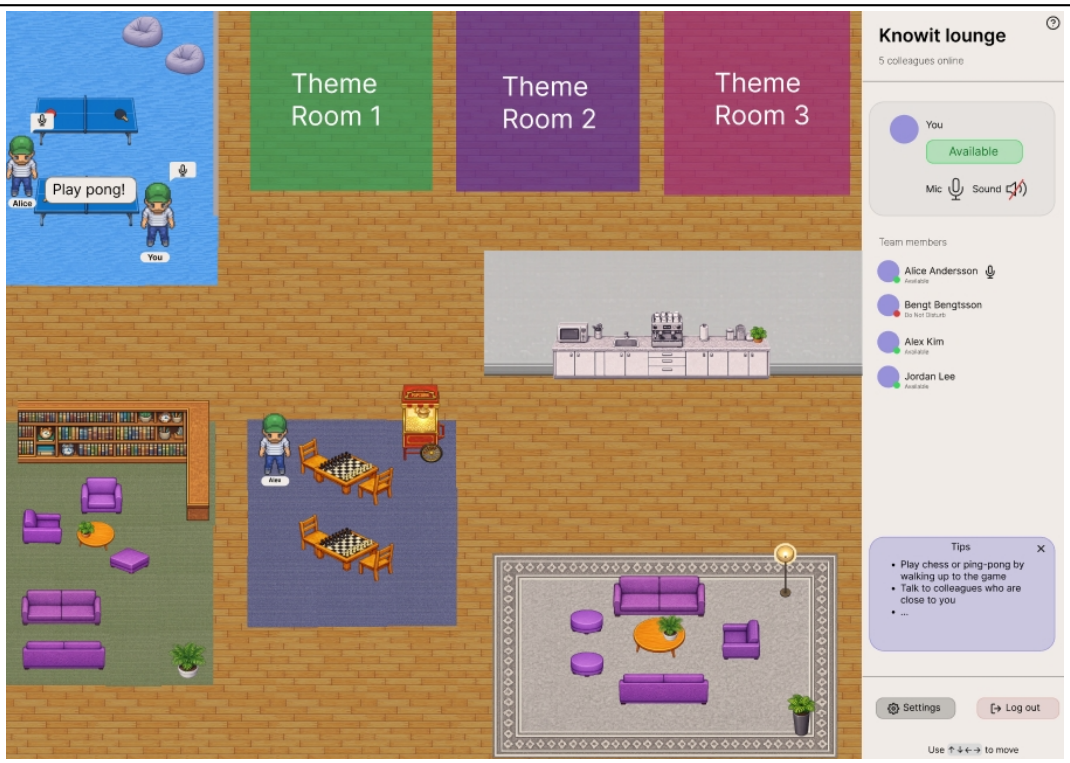


Figure 3 Overview of the second prototype iteration.

4.2.4 Guerrilla Evaluation

The revised prototype was briefly evaluated through guerrilla usability testing with one participant. The participant provided positive feedback and indicated that the implemented changes improved the overall usability and clarity of the interface. No major usability issues were identified, suggesting that the design revisions moved the prototype in the intended direction.

4.3 Phase 3: High-Fidelity Prototype and Evaluation

Following the iterative design work in Phase 2, a high-fidelity prototype was implemented to allow a more realistic evaluation of the proposed concept. This phase presents the implemented prototype together with the findings from the user evaluation.

4.3.1 High-Fidelity Prototype

The high-fidelity prototype builds on the design established in the Figma prototype, retaining the retro aesthetic and requirement-driven structure while introducing full interactivity. It consists of several views: a login page, a sign-up page, a user representation setup, a settings page, the main lounge, and pop-up windows for the two games.

The lounge (Figure 4) serves as the central interaction space, where users can move freely, see others' presence and status, and engage in conversation. A sidebar provides an overview of the current colleagues in the lounge, along with their availability status and whether their microphone is active. Users can also manage their own settings directly from the sidebar, including toggling their status between available and do not disturb, and adjusting their microphone and sound settings. At the bottom of the sidebar, a settings button allows users to edit their display name and appearance, or log out. A help button is also available, providing hints on what actions can be taken in the lounge.

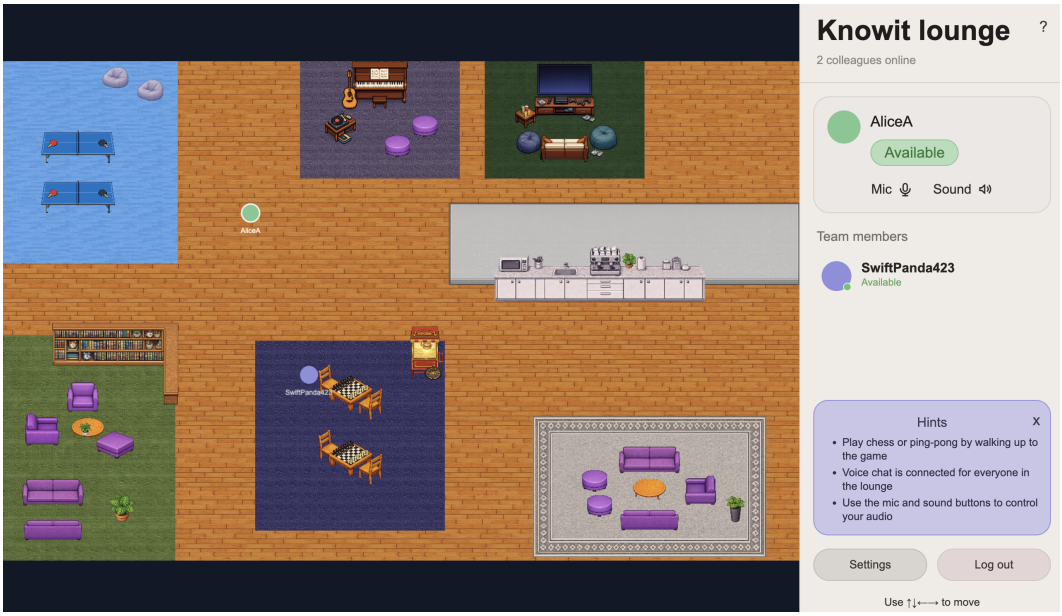


Figure 4 Overview of the lounge page of the high-fidelity prototype.

When creating an account, users are taken to a setup page before entering the lounge, where they can choose how they want to be represented, as shown in Figure 5.

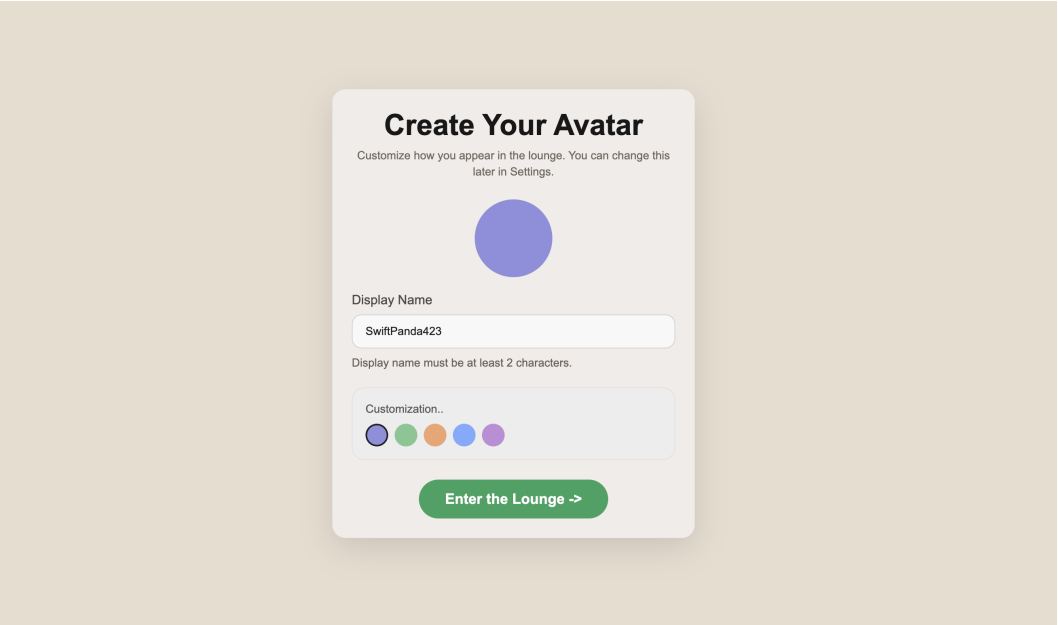


Figure 5 User representation setup interface used when creating an account.

Two games are available in the lounge. If two users choose to play chess, they are directed to a shared pop-up window where they can play together while communicating with each other. Pong is also available, though in this version users play against the computer. An example of the chess view is shown in Figure 6.

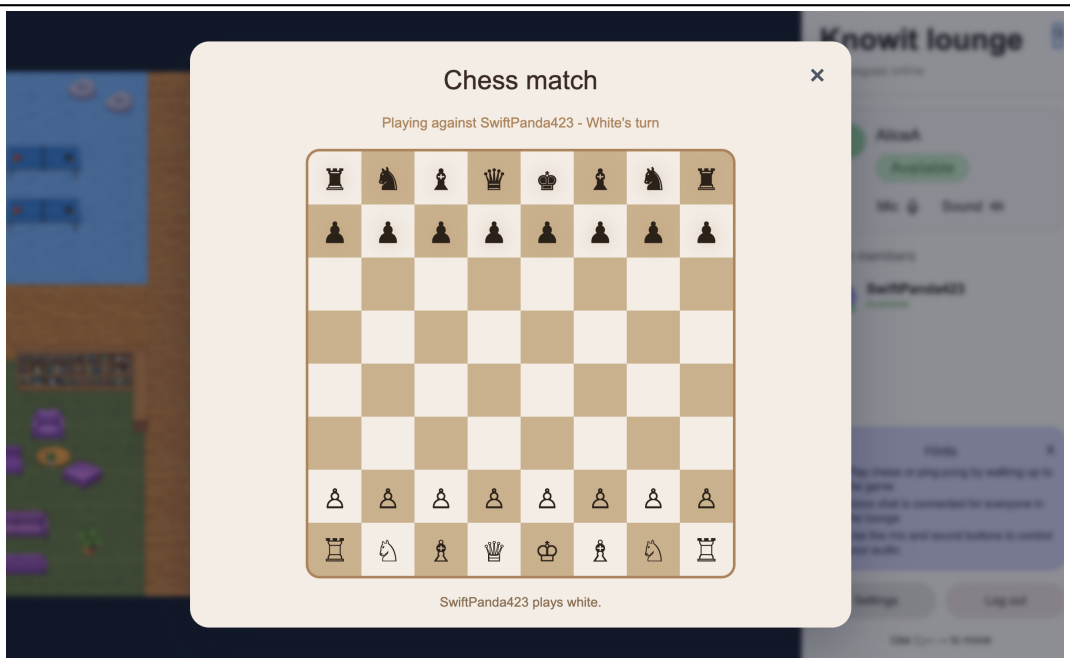


Figure 6 Example of a chess interaction within the lounge environment.

4.3.2 Think-Aloud Evaluation and Post-Test Interview

The think-aloud evaluation of the high-fidelity prototype identified several usability issues, primarily related to interaction discoverability and interface clarity. Avatar movement via the arrow keys was discovered quickly by all participants, however three out of four participants did not discover that Pong was controlled with the mouse. One participant overlooked the small on-screen prompt indicating how to launch the chess game. Another participant attempted to change their avatar color by clicking their profile in the sidebar rather than navigating to the settings panel, suggesting that the location of avatar customization was not immediately intuitive.

4.3.2.1 Findings from the High-Fidelity Prototype Evaluation

Beyond usability observations, the post-task questions revealed several recurring perceptions of the prototype as a social environment for informal interaction. Common patterns across participants' responses were identified and used to derive the findings presented below. These findings were analyzed in relation to the social and collaborative challenges identified in Phase 1.

Perceived Support for Spontaneous Interaction. The environment was generally per-

ceived as capable of supporting spontaneous interaction. Although some participants imagined using it in a more planned way, they generally felt that having a dedicated space removes the need to schedule a formal meeting. The lounge was described as laid-back and suitable for a digital coffee break or a quick game between tasks. These findings suggest that the lounge format can lower the threshold for spontaneous interaction, even if initial use may involve some degree of coordination.

Approachability and Reduced Social Threshold for Interaction. Approaching other users was experienced as easy and low-effort. The lounge was seen as particularly suitable for colleagues who are less comfortable initiating contact, as simply being present signals openness to interaction without putting pressure on anyone. This suggests that the spatial format supports a lower social threshold than direct messaging tools, where contact always implies a direct request for attention.

The Lounge as Less Transactional Than Existing Communication Tools. The platform was perceived as less transactional than tools like Microsoft Teams, where interactions tend to be question-and-answer exchanges with little room for casual conversation. Participant described interaction in Teams as direct requests for attention, whereas simply being present in the lounge communicated openness to conversation without pressure. This suggests that the lounge may fulfill a social function not currently supported by the communication tools used by the participants.

Shared Activities as Natural Entry Points for Interaction. The mini-games were seen as effective conversation starters by the participants and natural reasons to enter the lounge. Having a shared activity reduces the need for a work-related reason to reach out, making interaction feel more comfortable and spontaneous. These findings indicate that shared activities reduce the effort required to initiate informal interaction.

Visibility of Colleagues as a Social Cue. The sidebar displaying who is currently online was perceived as useful for identifying colleagues who are otherwise less visible in hybrid work, whether due to remote work or being on different projects. Participants compared it to the natural awareness of who is around in a physical office. While concerns about being monitored were raised, it was generally noted that colleagues and managers can already see each other in a physical office, meaning the lounge does not introduce a new form of visibility. This indicates that the visibility feature is experienced as a helpful social cue rather than a surveillance concern.

Conditions Influencing Adoption. Easy access was identified as a key condition for adoption, including staying logged in between sessions and having quick access via a bookmark. More broadly, adoption was seen as dependent on collective participation. If enough colleagues use it and there is something worth doing there, others are likely to follow. This suggests that the platform's value increases with the number of colleagues

actively using it, suggesting that organizational participation is as important for adoption as the platform itself.

4.3.2.2 User-Identified Improvement Areas

Participants suggested several improvements following the evaluation. Avatar movement speed was raised by one participant, who suggested adding a setting to control it. Two participants requested more personalized avatars, such as being able to upload a profile picture or customize their appearance, noting that the current placeholder circles made it difficult to identify who was who in the lounge. Making the chessboard larger was also suggested to make it more comfortable to play. These suggestions indicate a need for increased personalization, improved visibility, and refined interaction design. In particular, avatar customization and interface clarity appear to be important both for usability and user engagement.

4.4 Summary in Relation to the Research Questions

The evaluation suggests that the prototype successfully addressed the core design challenge identified in Phase 1: lowering the threshold for informal interaction in hybrid work. The post-task questions indicated that the prototype created a social atmosphere that participants experienced as distinctly different from conventional work tools. Compared to tools such as Teams and Zoom, which often require a specific reason for initiating contact, simply being present in the lounge signals openness and availability for interaction.

This suggests that the value of the prototype comes from the combination of visibility, shared activities, and an informal atmosphere, rather than from one specific feature alone. The sidebar showing who was online was seen as useful for knowing who was available to talk, reflecting how visibility supports social presence and makes colleagues feel more approachable [26]. The mini-games were seen as a natural way to start a conversation, supporting the idea that shared activities give people a reason to interact without a work-related agenda [27]. Together, the findings indicate that the functional requirements reflected relevant social and interaction-related needs in practice. These findings also align with the need for relatedness in SDT [28], as participants described the lounge as a place where social connections could naturally occur. However, participants noted that the platform would be useful only if colleagues were online at the same time and access was quick and easy, which is consistent with the adoption challenges reported in research on similar platforms [1].

Overall, these findings suggest that a 2D virtual workspace can support informal interaction by combining visibility, low-threshold interaction, and shared activities, while also requiring careful consideration of usability and organizational adoption.

5 Discussion

This section discusses the results in relation to the research questions and the aim of the thesis. The study contributes by showing that supporting informal interaction in hybrid work is not only a matter of providing communication tools, but of enabling social approachability, where colleagues feel present, reachable, and open to interaction with minimal effort. It further demonstrates how identified user needs, such as visibility, low-threshold interaction, and shared activities, can be translated into concrete design features perceived as meaningful by users. Together, these findings suggest that designing for informal interaction in hybrid work requires prioritizing social accessibility and spontaneous engagement alongside traditional collaboration support.

5.1 Designing for Informal Interaction in Hybrid Work

Both prior research and the interviews conducted in this study show that communication in hybrid work tends to become more planned, task-oriented, and socially limited than in physical office environments. This subsection discusses the results in relation to these challenges and what they suggest about how a 2D virtual workspace can be designed to support informal interaction in hybrid work. Taken together, the findings suggest that challenges to informal interaction in hybrid work stem less from a lack of communication channels and more from the perceived social effort involved in initiating contact. However, the conclusions drawn are based on a limited number of participants within a single organization, and should therefore be seen as indicative rather than generalizable.

A central finding concerns the significance of visibility and social presence. Participants reported that visibility of colleagues' presence and availability reduced uncertainty and made interaction feel more natural than communication through tools such as Microsoft Teams, where initiating contact was often perceived as interruptive or demanding. These observations align with Social Presence Theory, which proposes that visible social cues can make digital interaction feel more natural and approachable [26]. However, increased visibility may also introduce challenges. As discussed later in Section 5.3, making colleagues' presence and availability visible could cause concerns about surveillance or monitoring. The study's results show that while visibility is a key design element for supporting informal interaction, it needs to be balanced with user control. The status

feature, which allows users to set themselves as 'Do Not Disturb', partially addresses this concern, though additional options for managing visibility should be considered in future development.

The results also indicate that informal interaction in hybrid work is strongly linked to the perceived ease and naturalness of contacting colleagues. Several participants characterized traditional work tools as transactional and primarily task-focused, whereas the lounge environment was seen as more casual and social. Unlike tools that require scheduling meetings or direct requests for attention, the prototype enabled users to enter a shared space and interact spontaneously. This finding suggests that designing for low-threshold interaction is important for recreating the informal communication typical of physical workplaces. These results support previous research by Olson and Olson [27], which demonstrates that distributed work environments often reduce opportunities for spontaneous interaction and informal coordination. However, lowering the threshold for interaction does not guarantee that interaction will occur. As several participants observed, the platform would only be meaningful if colleagues actively used it. This suggests that organizational adoption may be as important as the platform design itself. At the same time, participants only interacted with the platform briefly, and it is therefore difficult to know whether the casual and social atmosphere would feel equally natural once the platform became a regular part of everyday work.

Another important finding was the role of shared activities in facilitating interaction. The mini-games were not only perceived as entertainment features, but also as natural conversation starters and shared social activities. Participants described how activities such as chess reduced the pressure of needing a specific work-related reason to contact someone. This suggests that shared activities may help create a shared context for interaction, as informal conversations naturally happen around shared activities or spaces in on-site offices. Therefore, the findings suggest that social interaction in digital work environments may benefit from designs that encourage lightweight and non-task-oriented engagement.

The findings can also be discussed in relation to Self-Determination Theory and the psychological need for relatedness [28]. Participants repeatedly described informal interaction as important for feeling connected to colleagues and maintaining a sense of belonging in hybrid work settings. The platform was therefore experienced not only as a communication tool but also as a social environment that supported connection and the workplace community. This suggests that digital environments designed for informal interaction can help enable social connection and belonging in hybrid work. It should, however, be noted that the prototype was tested in short sessions rather than in real everyday use, meaning that participants' responses primarily reflect first impressions rather than experiences grounded in long-term use. One factor that may have influenced participants' positive reactions is the novelty effect. Since the platform differs from

traditional workplace tools through its game-like and social design, participants may initially have experienced the environment as more engaging simply because it was new and different. Previous research in UX suggests that user experience can change over time as the novelty of a system decreases [18]. Therefore, whether the platform can sustain a feeling of connection and lasting engagement over time is something that needs to be explored through long-term deployment and future research.

In relation to the main research question, the findings suggest that a 2D virtual workspace can support informal interaction by combining visible presence, low-threshold opportunities for interaction, and shared social activities within a dedicated digital space.

5.2 Reflections on the Design Process

The iterative and user-centered nature of the design process shaped both the development of the prototype and the understanding of informal interaction in hybrid work. User involvement throughout the design process played an important role in grounding the prototype in employees' actual experiences of hybrid work rather than assumptions about how informal interaction occurs remotely. The early interviews were especially valuable in shaping the direction of the design, as they revealed that the challenge was not necessarily a lack of communication tools, but rather the social effort required to initiate informal interaction digitally. This influenced both the functional requirements and the overall design focus of the prototype.

Working iteratively also provided insight into how employees interpret and use digital social spaces. The think-aloud studies revealed usability issues that might otherwise have gone unnoticed, mainly regarding how participants interpreted different interface patterns and icons. This illustrated how easily designers can overestimate the intuitiveness of interface elements during development. Therefore, the changes made between the first and second iterations of the mid-fidelity prototype became important not only for usability, but also for reducing interaction barriers in a system intended to support informal communication. Small usability issues may appear minor during development, but can quickly become obstacles in tools meant to feel social and easy to engage with.

Conducting the study within an existing hybrid organization strengthened the contextual relevance of the findings, since participants could relate the prototype to established workplace routines and existing collaboration practices. This also influenced the type of insights generated, as employees described challenges shaped by movement between physical and digital work environments rather than fully remote work alone. The study most directly relevant to this work, Aoki *et al.* [1], focused on entirely remote software development teams and explicitly pointed out the need to examine how these environ-

ments function in different organizational contexts. By examining a hybrid organization specifically, where employees alternate between the office and home, the findings provide a more context-specific understanding of the social challenges of hybrid work than research on remote teams alone can offer.

A qualitative approach was particularly important for understanding how employees experienced and interpreted informal interaction in hybrid work settings. Many of the findings concerned perceived availability, hesitation, and the social effort involved in initiating contact, aspects that would have been difficult to capture through quantitative measures alone. The study did not focus on interaction frequency or usage patterns. Instead, interviews and qualitative evaluations were used to explore why employees experienced informal interaction as challenging and how they perceived the prototype in relation to their everyday collaborative practices. At the same time, future research could benefit from combining these qualitative findings with quantitative data. For example, usage logs could be used to track how often and for how long employees use the platform. Surveys could also measure changes in employees' sense of belonging and connection over time. Such data would complement the qualitative insights from this study and provide a more comprehensive understanding of how the platform influences informal interaction in practice.

Looking back at the process, there are aspects that could have been approached differently, given more time and resources. It should be noted that several participants took part in both the interviews and the prototype evaluations. While this ensured that participants understood the study's context, it may also mean they approached the prototype with a certain understanding of the problem it was designed to solve, which could have influenced how positively they experienced it. Participants with no prior involvement in the study may have evaluated the prototype differently.

Another factor worth considering is the limited time participants spent interacting with the prototype. Each evaluation session lasted about 20-40 minutes, where participants completed a small number of tasks and answered post-task questions. This is a very short exposure to a platform that is intended to be used as part of everyday work over time. Participants therefore had limited opportunity to form a fully considered opinion about how the platform would actually fit into their work routine, and their responses naturally reflect first impressions rather than deeper experience with the system.

Furthermore, since all participants were from the same organization and knew that the prototype was developed as part of a thesis project, there is a risk that they responded more positively than they would have otherwise. Participants may have felt inclined to respond positively, which could mean that doubts or negative reactions were not fully expressed. While the think-aloud method and open-ended questions were used to encourage honest feedback, the risk of this type of response bias can never be completely

avoided.

5.3 Ethical and Sustainable Aspects

This thesis presents the design and evaluation of a digital environment intended to support informal social interaction in hybrid work settings. Both the developed system and the research process involve important ethical and sustainability considerations.

From a system perspective, an important ethical aspect is privacy and visibility. The platform includes features such as presence indicators and shared virtual spaces. This makes users' availability visible to others, which could make spontaneous interaction easier, but it could also cause concerns about being monitored. As discussed in the background, increased visibility can support coordination, but could also make employees feel exposed [36]. It is therefore important that users can control their visibility and availability. Another important aspect is autonomy and voluntary participation. Although the platform is intended as a space for informal interaction, there is a potential risk that employees may feel pressure to use it. For example, employees may feel compelled to be present on the platform in order to stay socially connected at work. This highlights the importance of ensuring that participation remains genuinely voluntary.

In addition to these ethical considerations, the platform also raises questions related to sustainability and long-term well-being in hybrid work environments. From a sustainability perspective, the platform can have positive effects by supporting employee well-being and decreasing feelings of isolation in hybrid work. As discussed in previous research, informal interaction plays an important role in building relationships, engagement, and a sense of belonging [6, 21, 28]. Supporting these interactions digitally can therefore strengthen social connections within organizations. At the same time, there are possible negative effects: adding another digital platform may contribute to digital overload and make it harder to separate what is seen as work, a break from work, or free time, especially in remote settings. Therefore, it is important that the platform is simple, not intrusive, and clearly separated from formal work tasks. Additionally, it is important that the company using it has clear rules for platform use.

In addition to sustainability considerations, accessibility and inclusive design are also important aspects of the platform. Since the platform is still a prototype, several accessibility features, such as support for screen readers, color vision deficiencies, and other accessibility-related needs, have not yet been implemented. This could result in reduced participation and user exclusion, since the platform may not be equally usable. Improving accessibility should therefore be an important focus in future development. Following recognized guidelines, such as the Web Content Accessibility Guidelines (WCAG),

would help create an inclusive and more sustainable digital work environment.

Finally, ethical considerations were also relevant in relation to the research process itself. Ethical aspects were taken into account during the data collection in accordance with the ethical principles for research [37]. This means that participants in interviews and user evaluations were informed about the study, how their data would be used, and that taking part was voluntary. They gave informed consent before participating and could withdraw at any time. The collected data were handled confidentially and used solely for the purposes of this thesis.

5.4 Limitations

While the study provides insights into designing for informal interaction in hybrid work, several limitations should be considered when interpreting the findings. This study was deliberately limited in several ways to remain focused and manageable within the scope of the thesis. The project was carried out within an organization, Knowit Connectivity, rather than across multiple organizations. This decision was made to gain a better understanding of what is typical of an employee's day-to-day work experience in a hybrid work environment. As a result, the findings reflect the specific context of this organization and cannot be directly generalized to other organizations with different cultures or work practices.

A qualitative methodology was used throughout the study, quantitative measures such as interaction frequency and usage patterns were not collected. This was an intentional decision given the exploratory nature of the research question. However, it means that no statistical claims can be made based on the findings. The evaluation focused on participants' initial experience with the prototype rather than long-term use or adoption. Studying how the platform would be used over time was outside the scope of this project, and the findings therefore reflect first impressions rather than conclusions about sustained use in everyday hybrid work.

Lastly, the platform was designed more as a proof of concept and a first version, not as a completed product. Features like proximity-based voice communication and accessibility support were not included due to timing constraints. In particular, the absence of proximity-based audio limited the platform's ability to replicate smaller spontaneous walk-up interactions similar to those described by Aoki *et al.* [1]. The platform was also hosted locally during evaluation. Therefore, the evaluations represent participants' experience with the high-fidelity prototype, not a final product. These limitations should therefore be considered when interpreting the findings, particularly regarding the transferability and long-term applicability of the results.

6 Conclusions

This thesis explored how a 2D virtual workspace can be designed to support informal social interaction in hybrid work settings. The rise of hybrid and remote work has made organizations increasingly dependent on tools such as Microsoft Teams and Slack for daily communication. Although these platforms support structured, task-based collaboration, they provide limited opportunities for spontaneous, unplanned exchanges that typically occur in shared physical environments. Without dedicated spaces for informal interaction, hybrid workers risk losing the spontaneous exchanges that build relationships, strengthen team unity, and sustain a sense of community at work.

To explore this, a user-centered design process was used in collaboration with employees at Knowit Connectivity, involving interviews, iterative prototyping, and user evaluation. This enabled grounding the design in actual workplace experiences and improving the prototype based on user feedback. The study contributes both empirical insight into how employees experience informal interaction in hybrid work and practical knowledge about how those experiences can inform the design of digital social environments.

The results point to several design elements that matter for supporting informal interaction digitally. Making colleague presence visible, providing a drop-in space, and offering lightweight shared activities such as mini-games all shaped an environment that participants found more socially natural than conventional work tools. Rather than requiring a reason to initiate contact, the lounge allowed interaction to arise on its own. Together, these elements demonstrate that supporting informal interaction digitally is not only about providing a shared space, but about designing for social approachability by making colleagues feel visible, reachable, and open to spontaneous interaction.

That said, the study also shows that good design is not enough on its own. Whether a platform like this would actually be used depends on collective participation, organizational culture, and how well it fits into existing work routines. The prototype developed in this study should therefore be seen as a proof of concept rather than a finished product. Within this context, it demonstrates that a dedicated digital social space is both feasible and well-received, while also pointing to the need for further development and research.

6.1 Future Work

During the course of this thesis, several directions for future work were identified. These are divided into design and technical improvements to the prototype and suggestions for further research.

6.1.1 Design and Technical Improvements

The platform was developed as a first version, with several points for improvement identified during the thesis. The customization of avatars was frequently mentioned by the participants during the final evaluation of the high-fidelity prototype. The current prototype represents users as simple colored circles, making it difficult to identify colleagues in the lounge. Allowing users to upload profile pictures or further personalize their avatars could strengthen social presence within the environment.

The range of activities available in the lounge could also be expanded. The current prototype includes chess and a single-player version of Pong, and adding multiplayer functionality to Pong and introducing more shared activities would give users more reasons to spend time in the space and more natural entry points for conversation. Seasonal or event-based features, such as themed decorations or activities tied to holidays or office events, could also help keep the environment feeling fresh and engaging over time. More interactivity within the lounge itself, such as interacting with objects in the environment, could further contribute to a social atmosphere. These examples illustrate several potential directions for further development.

Another important improvement is the implementation of proximity-based audio, which was part of the design intent but could not be realized due to time constraints. Currently, all users in the lounge share a single voice room, regardless of their location. Adding proximity-based communication in the lounge could better replicate spontaneous conversations happening in physical office environments and likely improve support for informal interactions.

Lastly, the platform is currently hosted locally, which means it can only be accessed during controlled sessions rather than as part of everyday work. For the platform to function in a real organizational setting, it would need to be hosted online so that employees can easily access it from their work computers. Other important future developments relate to user account management, where practical features are currently missing, such as password recovery and account deletion. Accessibility improvements should also be addressed, including screen reader compatibility and support for color vision deficiencies, to support a more inclusive and accessible user experience for employees with different needs and abilities.

6.1.2 Further Research

The findings of this study also point to several directions for further research. The evaluation in this study focused on participants' first impressions of the platform rather than

their experience of using it over time. A natural next step would therefore be to study how the platform is actually used within a team over an extended period. This would provide stronger insight into whether the platform can sustain informal interaction over time and contribute to employees' sense of connection and belonging in hybrid work settings.

Since this study was conducted within a single organization, evaluating the platform in other organizational contexts with different workplace cultures and hybrid work practices could strengthen the transferability of the findings. This would help establish whether the findings and design principles from this thesis hold in other settings beyond Knowit Connectivity. Future research could also complement the qualitative findings of this study with quantitative methods, for example, by tracking usage patterns such as frequency and duration of platform use, or by using surveys to measure whether it affects employees' sense of belonging or connection over time. Together, these directions highlight the need for both continued design development and broader empirical evaluation to fully understand the role of digital environments in supporting informal interaction.

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A Appendix

A.1 Interview Guide

Semi-structured Interview Protocol

- ➢ 7 employees
- ➢ Time: around 40 min

Intro - Go to Participant Information Sheet

- Brief presentation of the study purpose.
- Confidential agreement
- Consent voice recording

Opening question:

1.Could you briefly describe your role at Knowit? How often do you work remotely and how often do you work at the office?

Informal Interaction

"Now I want to ask you about informal interactions at the workplace. By informal interaction, I mean work-related social conversations that are not part of formal meetings or structured tasks. For example unplanned talks in the corridor or by the coffee machine."

2. What role does informal interaction play in your work life?
3. How do informal interactions typically occur when you are at the office?

Probes:

- *Are there specific situations where informal interaction happens naturally?*
- *Can you describe a recent situation where informal interaction was important?*

4. When working remotely/in digital work environments, are there forms of informal interaction that you feel are lacking?

Probes:

- *Coffee breaks?*
- *Quick check-ins?*
- *Casual conversations before or after meetings?*
- *Shared small talk or humor?*

Social Presence when working remote

5. When you work remotely, how would you describe your sense of connection to your colleagues?

Probes: Can you tell me more about that? What influences that feeling? Does it vary depending on the situation or work tasks?

6. How do you experience accessibility or availability of colleagues when you work remotely?

Probes: What makes someone feel accessible? Can you give an example? How does that affect your day? How does that affect how you plan your day?

7. When you think about working at the office versus working remotely, what feels different socially?

Probe: I've heard some people say they miss the feeling of others being "around." What are your thoughts on that?

Design-Oriented Questions

"I would now like to explore your thoughts on a possible digital platform designed specifically for informal social interaction in hybrid work."

8. If there were a 2D virtual platform, like a digital lounge, how do you think you would use it?

Probes:

- *In what situations would you use it?*
- *Would it be spontaneous or planned?*
- *Would you use it alone or with specific colleagues?*
- *How often do you think you would use it?*
- *Would it replace something you already do today?*

9. What do you think such a platform should include to feel meaningful and worth using for you?

Probes:

- *Games? If so, what kind?*
- *Drop-in conversations?*
- *Small group spaces?*
- *Themed rooms (coffee, lunch, hobbies)?*
- *Shared activities?*
- *The ability to see who is online?*
- *Avatars or visual representation?*
- *Status indicators (e.g., "open for chat")?*

10. What would make you actually want to use this platform?

Probes:

- *What would make it fun?*
- *What would make it feel natural?*
- *What would make it easy to use?*
- *What would make it better than just using Teams/Slack?*

11. Is there anything that would make you avoid using such a platform?

Probes:

- *Lack of time?*
- *Too playful or not professional enough?*
- *Feeling monitored?*
- *Too complicated?*
- *Too many digital tools already?*
  Social pressure to participate?

Closing Questions

12. Is there anything else you would like to add regarding informal interaction in hybrid work?

Would it be okay if I reached out if I forgot to ask something or needed clarification?

Thank participant.

A.2 High-Fidelity User Evaluation Tasks and Questions

Hi-Fi Prototype - User Evaluation

- 4 employees
- Time: around 30 min.
- Brief presentation of the study purpose.
- Consent voice recording.

Tasks:

- Create an account and enter the lounge.
- Explore the lounge and describe what you think you can do there.
- Find a game in the lounge and try to start playing it.
- Change your display name or availability status.

Questions:

Question ID		Related Theme(s)
Q1	Does the environment feel like a place where you could naturally “drop in” spontaneously?	1, 2
Q2	Does it feel easy or difficult to approach other users in the environment?	2, 3
Q3	Does initiating contact here feel easier than via tools like Teams?	2
Q4	Do the activities feel like natural social entry points?	4
Q5	Does the visibility of others feel helpful or intrusive?	3
Q6	What would it take for you to actually want to use a platform like this with colleagues?	5

If there’s one thing you would change or add to the platform, what would it be?